

A Target-Centric Survey of Quantization-Aware Training

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Abstract

The rapid development of LLMs incurs prohibitive memory footprints and intensive computational demands. Quantization-Aware Training (QAT) techniques have emerged as a promising solution to address these challenges by explicitly *simulating quantization effects during model training*, yielding low-bit models that achieve accuracy comparable to their full-precision counterparts. In this work, we provide a target-centric survey of QAT, aimed at clarifying both its theoretical foundations and its evolving implementation landscape. We systematically review existing QAT methods through a target-centric taxonomy and synthesize cross-target differences in error characteristics, numerical formats, and strategy transferability. We further summarize QAT evaluation paradigms and discuss challenges in optimization and deployment, outlining potential directions for future research.

1 Introduction

Recent years have witnessed rapid development progress of large-scale foundation models capable of communicating across different languages, understanding diverse data modalities, and generating multimedia assets. The rapidly growing scale of these models demands intensive memory and computational resources far exceeding the capacity of most current edge devices, sometimes even consumer-grade GPUs (Nagel et al., 2021; Wei et al., 2024; Yang et al., 2025; Liu et al., 2025a). To mitigate these resource bottlenecks, model quantization serves as a fundamental technique, compressing the size of these massive models by lowering the bit-width of their parameters (Jacob et al., 2018; Nagel et al., 2021; Wei et al., 2024).

Model quantization methods can be broadly divided into Post-training quantization (PTQ) and

Quantization-aware training (QAT; Nagel et al. (2021); Wei et al. (2024); Yang et al. (2025)). PTQ applies low-precision quantization to a pre-trained model without further training to reduce model size and improve inference efficiency (Xiao et al., 2023; Liu et al., 2024b; Zhang and Shrivastava, 2024). However, directly quantizing a pre-trained model introduces significant performance degradations due to the limited representational range of low-bit weights. For instance, Bondarenko et al. (2021) identified that activations in residual connections contain structured outliers with high dynamic ranges, which are difficult to represent in low-bit formats. This has been repeatedly observed across numerous quantization methods; researchers are then cautious about directly applying PTQ.

In contrast, *QAT simulates quantization during training* using fake quantization modules, optimizing the model directly under quantized inference constraints, and often recovering the accuracy loss of PTQ, especially at ultra-low bit-widths such as INT4 (Jacob et al., 2018; Bengio et al., 2013). Recent QAT methods further combine quantization with parameter-efficient adaptation or distillation, making them increasingly applicable to large-scale LLMs (Xu et al., 2024; Chen et al., 2025).

There exist surveys covering QAT, however, they often discuss QAT as a component of the broader topic of neural network quantization, leaving the target-specific design principles of QAT underexplored. Table 1 summarizes the scope and taxonomy of representative quantization surveys and highlights how our target-centric, QAT-focused perspective differs from prior work.

In contrast, we primarily focus on QAT and organize the literature accordingly. This perspective highlights how the numerical characteristics, lifetime, reuse frequency, and deployment constraints of different quantization targets shape the design of QAT schemes, and why targeting different modules in QAT results in diverse benefits and costs.

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